

# Machine Learning — Day 1 Notes

From AI basics → Data → Dataset → Features → Labels

## Day 1 Goal

By the end of Day 1, you should understand what Artificial Intelligence (AI) and Machine Learning (ML) are, what data and datasets are, and the difference between features and labels.

## 1. What is Artificial Intelligence (AI)?

**Artificial Intelligence is the field of making computers perform tasks that normally require human-like intelligence.**

Examples: understanding speech, recognizing images, recommending videos, answering questions, and making decisions.

**Remember:** AI is the broad field. Machine Learning is one way of building AI systems.

## 2. What is Machine Learning?

**Machine Learning is a way of teaching computers to learn patterns from data and use those patterns to make predictions or decisions.**

Traditional programming usually works like: **Rules + Data → Output.**

Machine Learning works more like: **Data + Examples → Learning → Model → Prediction.**

Example: Instead of manually writing every rule for detecting spam emails, we can give an ML system many examples of spam and non-spam emails so it can learn patterns.

## 3. What is Data?

**Data is information or facts about something.**

Data can be numbers, text, categories, images, audio, video, dates, measurements, and more.

**Example:** A student's age is 18, study time is 4 hours, and attendance is 80%. These are pieces of data.

## 4. What is a Dataset?

**A dataset is an organized collection of data, usually containing many examples.**

| Age | Study Hours | Attendance | Result |
|-----|-------------|------------|--------|
| 18  | 2           | 60%        | Fail   |
| 19  | 5           | 85%        | Pass   |
| 18  | 1           | 50%        | Fail   |
| 20  | 6           | 90%        | Pass   |

The entire table is a **dataset**. Each row is one example, and each column contains one type of information.

**Easy analogy:** One piece of information = data. A collection of organized examples = dataset.

## 5. What is a Feature?

A **feature** is an **input variable** or **piece of information** that is used by a **Machine Learning** model to **make a prediction**.

Think of features as **clues** given to the model.

In the student dataset above, if we want to predict the result, possible features are **Age, Study Hours, and Attendance**.

Features can be numbers, categories, text, images, audio, or other useful information. In many ML systems, raw information is converted into a form the model can process.

## 6. What is a Label?

A **label** is the **known answer** or **target** that we want the **Machine Learning** model to **learn to predict**.

In the student example, **Result** is the label because it is the answer we want the model to predict: Pass or Fail.

## Feature vs Label

| Concept | Meaning                          | Student Example              |
|---------|----------------------------------|------------------------------|
| Feature | Input / clue used for prediction | Age, Study Hours, Attendance |
| Label   | Answer / target to predict       | Result                       |

## 7. Important Example

Suppose we want to predict a house's price.

| Area       | Bedrooms | Location  | Price    |
|------------|----------|-----------|----------|
| 1200 sq ft | 2        | Hyderabad | ■60 lakh |
| 1800 sq ft | 3        | Hyderabad | ■90 lakh |

**Features:** Area, Bedrooms, Location

**Label:** Price

## 8. Day 1 Mental Map

AI → Machine Learning → Data → Dataset → Features + Labels

The next stage will be understanding how an ML algorithm learns from features and labels to create a model.

## 9. Practice Task

**House Price Prediction:** A real-estate company wants to predict the price of a house using its area, number of bedrooms, and location. Identify the data, dataset, features, and label.

### Day 1 — Key Definitions

- **AI:** The broad field of making computers perform tasks requiring human-like intelligence.
- **Machine Learning:** A way of learning patterns from data to make predictions or decisions.
- **Data:** Information or facts.
- **Dataset:** An organized collection of data/examples.
- **Feature:** An input piece of information used to make a prediction.
- **Label:** The known answer/target that the model is meant to predict.